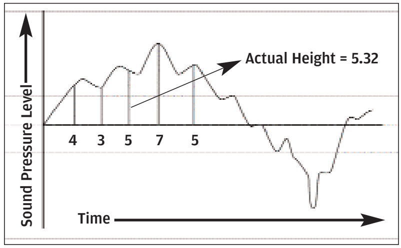
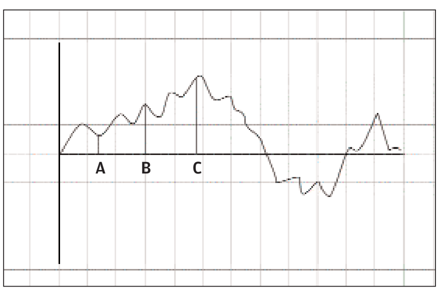


above represent? It can represent the voltage, or sound pressure level, or loudness, of that frequency. Essentially, the vertical level of the wave represents *how much of this frequency is present in the music.* So, we "look" at the wave continuously—several times per second—and note the value (the height). Each time we look at it, we represent the height by an approximation. This is called quantisation.

In the image above, the actual height may be 5.32, but we represent it as 5. The 0.32 is the "quantisation error."

So where does the "digital" come in? It's simple: that "5" is represented digitally, as "101". (If it were indeed "5".) Now, the more the number of levels we define for our representation, the more accurate our representation will be. Say we have only 10 levels, from 1 to 10. Then 5.3 would be coded as 5, 6.8 would be coded as 7, and so on. But say we have 20 levels. Then 5.3 would be coded as 5.5, 6.3 would be coded as 6.5, and so on. We're getting more accurate. Now, representing 10 levels only needs four bits; representing 20 levels requires five bits.



We can now define the sampling rate and bitrate. The sampling rate (the "44.1 kHz" you see in Winamp or any media player when a clip plays) is the number of times the waveform is "looked at" per second. A low sampling rate means we only look at the waveform at points A, B, and C above.